

A nonzero projection with scalar corner

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_nonzero_projection_scalar_corner`

theorem

Constructs a projection in the original algebra from an isolated non-scalar character of a unitization subalgebra.

Statement

Let A be a nonzero complex C^* -algebra, with no unit assumed. Let H be a separable complex Hilbert space, and let $\pi : A \rightarrow \mathcal{B}(H)$ be a nonzero irreducible representation representing the unique unitary-equivalence class of nonzero irreducible representations of A . There is $p \in A$ with

$$p^* = p = p^2, \quad p \neq 0, \quad \forall a \in A \exists c \in \mathbb{C} : pap = cp.$$

Assumptions

Separability is imposed on H . The algebra need not be unital or separable. The scalar c may depend on a , while the projection p is fixed.

Conclusion

The same nonzero star projection has a one-dimensional algebraic corner $pAp = \mathbb{C}p$.

Proof route

Find an isolated character away from the scalar character, take its characteristic-function projection, show that its scalar coordinate vanishes, and extend its scalar corner from the abelian subalgebra to the ambient algebra.

▼ Proof steps

1. Write $\tilde{A} = \mathbb{C} \oplus A$, $\iota(a) = (0, a)$, and $\chi_\infty(\lambda, a) = \lambda$. Choose $a_0 \neq 0$ in A and put $b = a_0^* a_0 \neq 0$. In the unitization $\tilde{A}$, its image $\iota(b)$ is self-adjoint. Apply Maximal abelian containment to obtain a maximal abelian unital $*$ -subalgebra D containing $\iota(b)$. It is closed and commutative. The character countability theorem Countability of this character space applies to the given singleton π , separable H , and this closed D .
2. The element $\iota(b) \in D$ is nonzero and has scalar coordinate zero. Together with countability, these are the inputs to An isolated non-scalar character, yielding an isolated character $\chi \neq \chi_\infty|_D$. Its singleton is both open and closed in the character space. Thus the continuous function $1_{\{\chi\}}$ corresponds under the Gelfand transform to a nonzero projection $q \in D$ satisfying

$$qd = \chi(d)q \quad (d \in D),$$

by The projection of an isolated character.

3. That characteristic function is zero at the different character $\chi_\infty|_D$, so $\chi_\infty(q) = 0$. This value is precisely the scalar coordinate of $q \in \tilde{A}$. Therefore $q = \iota(p)$ for an element $p \in A$. Injectivity of ι transfers $q^* = q = q^2$ and $q \neq 0$ to $p^* = p = p^2 \neq 0$.
4. The ambient corner calculation is From maximal abelianness to a scalar ambient corner. For any $z \in \tilde{A}$, put $x = qzq$. Since D is commutative and $qd = dq = \chi(d)q$, for $d \in D$ we get

$$xd = qzqd = \chi(d)x = dqzq = dx.$$

Maximal abelianness therefore puts x in D . Apply the earlier equation to this x : $qx = \chi(x)q$. But $q^2 = q$ makes $qx = x$, so $qzq = \chi(x)q$.

5. Finally set $z = \iota(a)$ for an arbitrary $a \in A$. With $q = \iota(p)$, the last equation is $\iota(pap) = \iota(cp)$ for $c = \chi(q\iota(a)q)$. Injectivity of ι gives $pap = cp$. This proves the universal scalar-corner condition on the same p .

Main citations

- Exact theorem and proof
- Maximal abelian containment
- Countability of characters
- Isolation away from the scalar character
- The projection of an isolated character
- From maximal abelianness to a scalar ambient corner

Lean source signature (exact)

```
theorem exists_nonzero_projection_scalar_corner [Nontrivial A]
  [TopologicalSpace.SeparableSpace H]
  (pi : NonUnitalCStarRepresentation A H)
  (hsingle : IsSingletonIrreducibleModel.{u, v, u} pi) :
  ∃ p : A, IsStarProjection p ∧ p ≠ 0 ∧
    ∀ a : A, ∃ c : ℂ, p * a * p = c • p
```

In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
[TopologicalSpace.SeparableSpace H]	Separability of the representation space, used to count characters.
(pi : NonUnitalCStarRepresentation A H)	The specified $*$ -representation $\pi : A \rightarrow \mathcal{B}(H)$; no unit-preservation equation is required.

In the source	Mathematical meaning
<code>(hsingle : IsSingletonIrreducibleModel.{u, v, u} pi)</code>	The specified representation is nonzero irreducible, and every nonzero irreducible $*$ -representation of the same algebra is unitarily equivalent to it.
<code>∃ p : A</code>	One element is obtained in A itself, after the unitization projection has zero scalar coordinate.
<code>IsStarProjection p</code>	That element satisfies both $p^* = p$ and $p^2 = p$.
<code>p ≠ 0</code>	The same projection is nonzero.
<code>∀ a : A, ∃ c : ℂ, p * a * p = c • p</code>	For each algebra element a choose a complex scalar c , possibly depending on a , so that the entire compressed product pap is cp .

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_nonzero_projection_scalar_corner`

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